

Screenshots for Set Up SSH Authentication on a Linux Server

```
(kali㉿kali)-[~/Desktop]
$ sudo apt update
Hit:1 http://http.kali.org/kali kali-rolling InRelease
2155 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Updating all packages

```
(kali㉿kali)-[~/Desktop]
$ sudo apt install ufw -y
ufw is already the newest version (0.36.2-10).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2155
```

Downloading the latest version of the firewall

```
(kali㉿kali)-[~/Desktop]
$ sudo ufw enable
Firewall is active and enabled on system startup
```

Activated the firewall

```
(kali㉿kali)-[~/Desktop]
$ sudo systemctl enable --now ssh
[sudo] password for kali:
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink '/etc/systemd/system/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
```

Enabling SSH on computer

```
(kali㉿kali)-[~/Desktop]
$ sudo ufw allow ssh
Rule added
Rule added (v6)
```

Allowing SSH through the firewall

```
(kali㉿kali)-[~/Desktop]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:70:3c:dd brd ff:ff:ff:ff:ff:ff
    inet 10.200.0.91/24 brd 10.200.0.255 scope global dynamic noprefixroute eth0
        valid_lft 676691sec preferred_lft 676691sec
    inet6 fe80::f418:5177:346a:105c/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali㉿kali)-[~/Desktop]
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=19.9 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=21.3 ms
^C
 8.8.8.8 ping statistics —
 2 packets transmitted, 2 received, 0% packet loss, time 1002ms
 rtt min/avg/max/mdev = 19.927/20.614/21.302/0.687 ms
```

Pings and IP addresses are confirmed

```
[skyler@Skyler skyler]$ ssh kali@10.200.0.91
Linux kali 6.16.8+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-09-24) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Jul 22 14:58:48 2026 from 10.200.0.66
(kali㉿kali)-[~]
$
```

Remotely accessed the server from my local computer

```
[skyler@Skyler skyler]$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/skyler/.ssh/id_ed25519):
/home/skyler/.ssh/id_ed25519 already exists.
Overwrite (y/n)?
[skyler@Skyler skyler]$ █
```

SSH keys have been made

```
skyler@Skyler/mnt/c/Users/s X + -
[skyler@Skyler skyler]$ ssh-copy-id kali@10.200.0.91
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/skyler/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
kali@10.200.0.91's password:
Connection closed by 10.200.0.91 port 22
[skyler@Skyler skyler]$ █
```

SSH keys have been copied to the remote server

```
[skyler@Skyler skyler]$ ssh kali@10.200.0.91
Linux kali 6.16.8+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-09-24) x86_64

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└─(kali@kali)─[~]
└─$ █
```

SSH is now possible without password confirming that the keys are working